

The effect of half-shaft torsion dynamics on the performance of a traction control system for electric vehicles

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Abstract

This article deals with the dynamic properties of individual wheel electric powertrains for fully electric vehicles, characterised by an in-board location of the motor and transmission, connected to the wheel through half-shafts. Such a layout is applicable to vehicles characterised by significant power and torque requirements where the adoption of in-wheel electric powertrains is not feasible because of packaging constraints. However, the dynamic performance of in-board electric powertrains, especially if adopted for anti-lock braking or traction control, can be affected by the torsional dynamics of the half-shafts. This article presents the dynamic analysis of in-board electric powertrains in both the time domain and the frequency domain. A feedback control system, incorporating state estimation through an extended Kalman filter, is implemented in order to compensate for the effect of the half-shaft dynamics. The effectiveness of the new controller is demonstrated through analysis of the improvement in the performance of the traction control system.

Keywords

In-board electric powertrain, half-shaft torsional compliance, traction control

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Introduction

The traditional layout of electric vehicle powertrains consists of a central motor, a single-speed or multiple-speed transmission, a differential and half-shafts.^{1–4} However, the future of electromobility is likely to be based on the adoption of individually controlled powertrains for each wheel.⁵ From the viewpoint of vehicle dynamics, this solution provides significant advantages, because of the possibility of implementing direct yaw moment control for enhancement of the vehicle response in steady state and transient conditions,^{6–8} and for actuation of the anti-lock brake system (ABS) and traction control (TC) functions. This is achievable through the precise control of the electric powertrain torque at the wheel. However, current regenerative braking systems for electric powertrains usually leave the modulation of the braking torques to the friction brakes and reduce the regenerative action during ABS cycling,^{9–11} while TC systems based on actuation of the internal combustion engines are characterised by a relatively low bandwidth. The low reaction time and high bandwidth of electric motor drives potentially allow the development of ABS controllers

based on direct slip control^{12–17} instead of conventional hybrid controllers based on the combination of wheel acceleration control and wheel slip control.¹⁸

Individual in-wheel electric powertrains provide an additional benefit from the viewpoint of the design of the overall vehicle chassis, by permitting an increase in the available volume in the area between the wheels. However, current motor technology is limited in terms of the power and the torque density, which makes in-wheel powertrains, with their motor drive unit incorporated into the unsprung mass of the vehicle, a viable solution for only small- and medium-size cars, with a relatively low performance. Despite recent studies^{19,20} which suggest a past overestimation of the importance of this factor, in-wheel electric powertrains are associated with an undesired increase in the vehicle's

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unsprung mass, with the subsequent ride comfort and vehicle dynamics implications (for rough-road conditions). Moreover, the adoption of an in-wheel powertrain on the wheels of a steering axle increases the turning radius of the vehicle, owing to the related packaging issues.

As a consequence, a possible solution for the implementation of individual electric powertrains, without being subject to the limitations of the in-wheel layout, is the implementation of in-board electric powertrains. These are characterised by a motor drive and a transmission connected to the vehicle chassis through a powertrain mounting system (possibly including bushings with non-linear elastic and damping properties), and by a half-shaft with constant-speed joints for driving the wheel. However, the main disadvantage of the in-board electric powertrain arises from the torsional dynamics of the half-shaft and the subsequent first torsional mode of the powertrain. The torsional dynamics of the electric drivetrain arise because the half-shaft can be considered as the combination of a torsion spring and damper (with a very low damping ratio) in parallel, connecting the equivalent mass moment of inertia of the drivetrain to the inertia of the wheel. Moreover, the torsional dynamics of the system are significantly affected by the slip ratio dynamics of the tyres, owing to the combination of the steady state non-linear characteristics and the relaxation length of the tyres. In a first approximation, the tyres can be represented as non-linear dampers within the system. Also the dynamics of the electric powertrain mounting system, especially if the drivetrain includes bushings, are relevant to driveability of the vehicle, because of the forces transmitted to the vehicle chassis by the powertrain mounting system. All these phenomena could interfere with the vehicle driveability, by affecting the jerk dynamics, as is well known for internal-combustion-engine-driven vehicles,²¹ usually characterised by a first natural frequency of the powertrain between 4 Hz and 7 Hz. The torsional mode of the powertrain could also reduce the effectiveness of ABS and TC actuation. However, the literature about electric vehicle control has not investigated this aspect yet.

This article provides a comprehensive analysis of the dynamics of in-board electric powertrains, in both the frequency domain and the time domain. A potential limitation in the actuation of effective ABS and TC systems is demonstrated. A feedback controller, based on an extended Kalman filter²² for estimation of the half-shaft torque, is implemented for compensation of the half-shaft torsional dynamics during ABS and TC intervention, which is a novel control approach for this important application. A similar algorithm has been implemented by Amann et al.²³ for compensation of the effect that the transmission has on the driveability of an electric vehicle.

The simulation model

This section describes the basic principles of the implemented non-linear model of the longitudinal dynamics of a case-study front-wheel-drive vehicle with individually controlled powertrains. A non-linear powertrain model is developed (and subsequently used as a simulator), from which a linearised vehicle model is derived for controller design purposes. The model presented in this section includes a significant number of simplifications and was selected as the simplest non-linear model capable of reproducing the dynamics of interest for the specific application.²¹

Figure 1 is the schematic diagram of an individual wheel powertrain, characterised by a three-stage single-speed transmission with a layshaft layout. The equivalent mechanical model implemented in this article consists of three degrees of freedom: the vehicle, the wheel and the powertrain. The half-shaft is modelled as a torsion spring and damper with two mass moments of inertia at its ends. The damping coefficient of the half-shaft is low owing to the very small internal damping of the material (steel). The connection between the wheel and the equivalent inertia of the vehicle is represented by the tyre. The angular speeds of the wheel and the equivalent inertia of the vehicle differ because of the slip ratio.

The electric motor reaction time and slew rate are included in the simulation model. The specific motor drive unit of this vehicle application is of a switched reluctance topology, characterised by a particularly low value of mass moment of inertia. A small time constant τ_M takes into account the effect of the electric dynamics of the motor (because of its equivalent circuit parameters) and is given by

$$\tau_M \dot{T}_M + T_M = T_{M,ref} \quad (1)$$

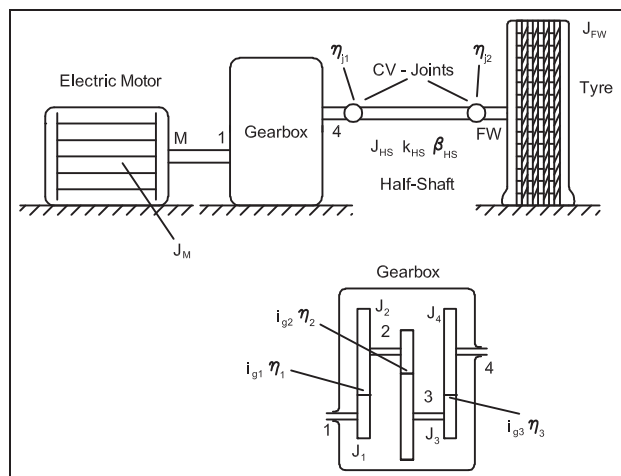


Figure 1. Schematic diagram of the individual in-board electric powertrain with a zoomed view of the three-stage single-speed transmission.

The overall powertrain dynamics from the motor shaft to the transmission output shaft are modelled by

$$T_M - T_{windage} - \frac{i_{g1}i_{g2}i_{g3}}{\eta_1\eta_2\eta_3\eta_{j1}} T_{HS} = \left[(J_M + J_1) + \frac{i_{g1}^2}{\eta_1} J_2 + \frac{(i_{g1}i_{g2})^2}{\eta_1\eta_2} J_3 + \frac{(i_{g1}i_{g2}i_{g3})^2}{\eta_1\eta_2\eta_3} \left(J_4 + \frac{1}{2} \frac{J_{HS}}{\eta_{j1}} \right) \right] \ddot{\theta}_M \quad (2)$$

where the half-shaft torque T_{HS} is given by

$$T_{HS} = \beta_{HS}(i_{g1}i_{g2}i_{g3}\dot{\theta}_M - \dot{\theta}_{FW}) + k_{HS}(i_{g1}i_{g2}i_{g3}\theta_M - \theta_{FW}) \quad (3)$$

A look-up-table-based model is adopted for estimation of the windage losses $T_{windage}$, which are functions of the rotor angular speed.

The transmission efficiency is modelled through a look-up-table-based model, as a function of the input torque, the speed and the operating temperature. Starting from T_{HS} , the net torque T_{FW} transmitted by the half-shaft to the wheel is calculated from

$$T_{HS} - \frac{T_{FW}}{\eta_{j2}} = \frac{J_{HS}}{2} \ddot{\theta}_{FW} \quad (4)$$

The rotational equilibrium of the driven wheel is given by

$$T_{FW} - T_{d,FW} - f_{r,FW}R_W = J_{FW}\ddot{\theta}_{FW} \quad (5)$$

The rolling resistance coefficient f_r of the tyres is modelled as a parabolic function of the tangential speed of the wheels. The longitudinal force $T_{d,FW}/R_W$ of the tyres is modelled using the Pacejka magic formula,²⁴ including a relaxation length model for the evaluation of the transient effects, according to

$$\frac{L_{tyre}}{\dot{\theta}_{FW}R_W} \dot{T}_{d,FW} + T_{d,FW} = T_{FW, Pacejka} \quad (6)$$

where L_{tyre} is a function of the wheel speed, the vertical load of the tyres and the slip ratio.

The vehicle's longitudinal dynamics are modelled through an equivalent moment of inertia, which is subject to equivalent torques due to the aerodynamic drag and the driven and undriven wheels (the rear wheels for the specific case-study vehicle of this article). The dynamics of the equivalent inertia of the vehicle are described by

$$2T_{d,FW} - 2f_{r,R}F_{z,R}R_W - \frac{1}{2}\rho C_d SR_W^3 \dot{\theta}_V^2 = (2J_{RW} + mR_W^2) \ddot{\theta}_V \quad (7)$$

Given the focus of the activity, the vertical load transfers of the wheels due to the longitudinal acceleration and deceleration levels were computed without

considering the effects of the suspension stiffness and the damping and without taking into account the vertical dynamics of the tyres. For example, the vertical load on each front wheel is given by

$$F_{z,FW} = mg \frac{b - u_{RW}}{2(L + u_{FW} - u_{RW})} + \frac{1}{4(L + u_{FW} - u_{RW})} \rho SR_W^2 [-C_d h_{CG} + C_m L - C_l(b - u_{RW})] \dot{\theta}_V^2 - m \frac{h_{CG}}{2(L + u_{FW} - u_{RW})} R_W \ddot{\theta}_V \quad (8)$$

A similar equation is adopted for the computation of the vertical load on the rear axle. The most relevant vehicle data are included in Table 3 of Appendix 3.

Linearization permits the analysis of the frequency response of the system and control system design. The model described by equations (1) to (8) was subjected to linearization around an operating condition, defined by values of the vehicle speed and the slip ratio. In the linear model, the tyre is considered as an equivalent torsion damper between the wheel and the vehicle inertia, where the equivalent damping coefficient is obtained through linearization of the Pacejka magic formula around the selected operating point. The main nonlinearities of the system are due to the force characteristics of the tyres. In particular, the absolute value of longitudinal slip stiffness of the tyres is positive and progressively decreases from the condition of zero slip until the peak value of longitudinal force, at which it assumes a null value. When the longitudinal slip stiffness of the tyres is null, the individual powertrain dynamics are completely decoupled from the dynamics of the equivalent inertia of the vehicle. This offers theoretical justification of the simplification carried out by most researchers (see, for example, the paper by Johansen¹²) in the derivation of the dynamic equation of the slip ratio within the implementation of the model-based ABS. The dynamics of the wheels are much more pronounced and at higher frequency values than the longitudinal dynamics of the vehicle for conditions of slip ratios around the value generating the peak longitudinal force of the tyres. Beyond the peak longitudinal force, the longitudinal slip stiffness of the tyres is negative. The linearised model was implemented in a state-space formulation

$$\begin{aligned} \dot{X} &= AX + BU \\ Y &= CX + DU \end{aligned} \quad (9)$$

The detailed mathematical formulation of the elements of the X and U vectors and the A and B matrices defined in equation (9) is included in Appendix 2, with some marginal additional simplifications (e.g. the windage losses are neglected). Additionally, it is possible to predict that most in-board implementations of electric powertrains will be usually characterised by a lower length of the half-shafts (in comparison with internal-combustion-engine-driven powertrains), owing to the

width of the system formed by the two electric motor units per each axle.

Assuming some particular parameters for the vehicle (Appendix 3), the frequency response of the model was analysed for a number of different linearization points, given the absence of any literature about the frequency response of such a powertrain layout.

Figure 2 contains the Bode plots of the adimensionalized half-shaft torque response to an input motor torque, for different values of the slip ratio (linearization point), and two values of the half-shaft length (i.e. half-shaft torsion stiffness), at a vehicle speed of 50 km/h, in conditions of a high friction coefficient ($\mu = 1$) between the tyre and the road surface. i_g^* is the overall single-speed transmission ratio ($i_g^* = i_{g1}i_{g2}i_{g3}$). At low values of the slip ratio (graphs at a null slip ratio in the figure), the frequency response is characterised by two peaks, at the natural frequencies of the system. The first is visible in the diagrams and is located between 80 rad/s and 90 rad/s (about 13 Hz and 14 Hz), while the second peak is not visible, as it is located beyond the frequency limits of the plot. When the value of the slip ratio at the linearization point is increased to 0.05, the low-frequency peak in the half-shaft torque response is subject to an amplitude reduction. Moreover, the frequency of the second resonance peak decreases, and the two peaks become closer. This is the range of slip ratios usually relevant for vehicle driveability. Finally, for values of the slip ratio close to that corresponding to the peak longitudinal force of the tyres, the system experiences a significant resonance peak between 100 rad/s and 200

rad/s (about 16 Hz and 32 Hz). The frequency range of ABS modulation in common systems based on hydraulic actuation of the friction brakes is between 4 Hz and 7 Hz. According to Murata,²⁵ the adoption of in-wheel motors can increase the ABS modulation frequency up to 9–10 Hz. This is the range of interest of this simulation model. Beyond these limits, there could be modal interactions with the longitudinal motion of the wheels relative to the chassis, owing to suspension compliance, usually in the region of 40–100 Hz.²⁶

Figure 3 plots the sensitivity of the frequency response of the same electric powertrain to different values of the tyre-road friction coefficient, for a relatively low slip ratio ($\sigma = 0.05$ in Figure 3(a)), relevant to vehicle driveability, and a slip ratio close to that corresponding to the peak longitudinal force of the tyres ($\sigma = \sigma_{PEAK} - 0.01$ in Figure 3(b)). The frequency response of the system is strongly dependent on μ , with the only exception being represented by the case of system linearization at exactly σ_{PEAK} , where all the Bode plots (irrespective of μ) are coincident. In fact, in that condition the wheel dynamics are decoupled from the vehicle dynamics, and so the longitudinal slip stiffness of the tyres is equal to zero. In the analysis of this preliminary research activity, the variation in the shape of the force characteristic of the tyres as a function of μ was neglected (all the characteristics were scaled down according to the assigned friction coefficient) and the existence of the peak longitudinal force of the tyres was considered also in conditions of very low friction. Future analysis will have to investigate the frequency

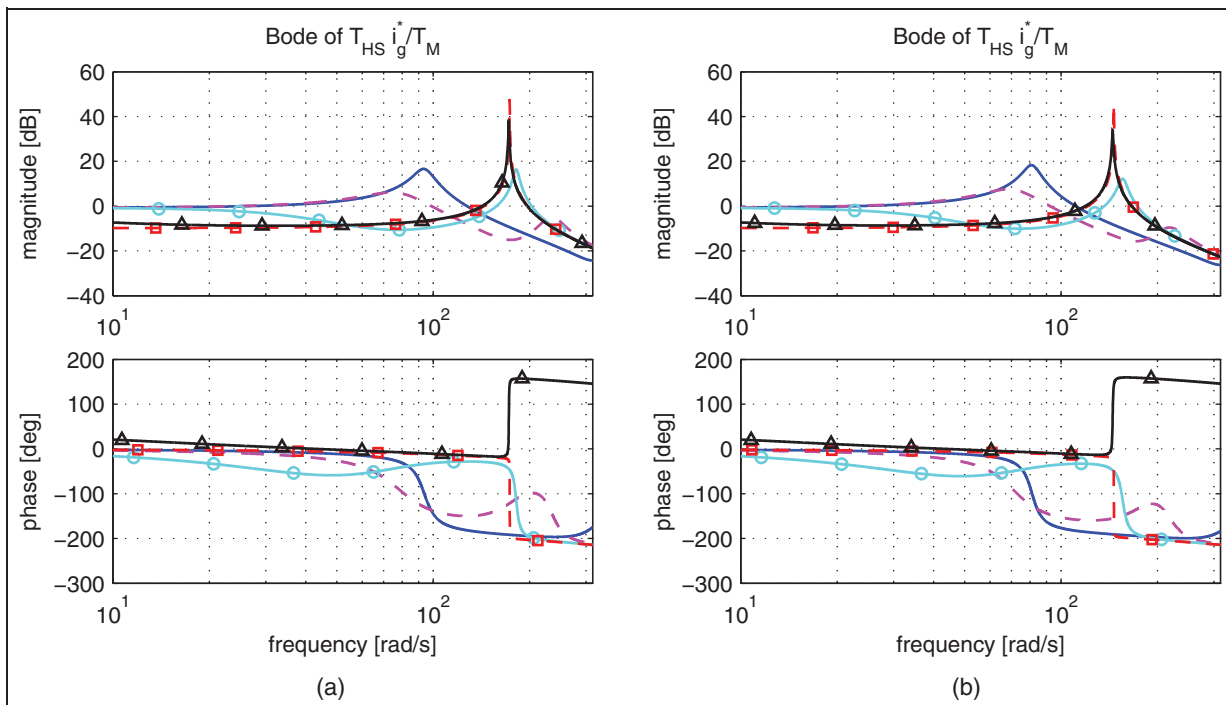


Figure 2. Bode plots (magnitude and phase) of the adimensionalized half-shaft torque for an input torque request and half-shaft lengths l_{HS} of (a) 506 mm and (b) 360 mm, at different values of the longitudinal slip ratio of the tyres (vehicle speed, 50 km/h): —, 0.0; ---, 0.05; ○, 0.1; □, 0.138 (peak); △, 0.15.

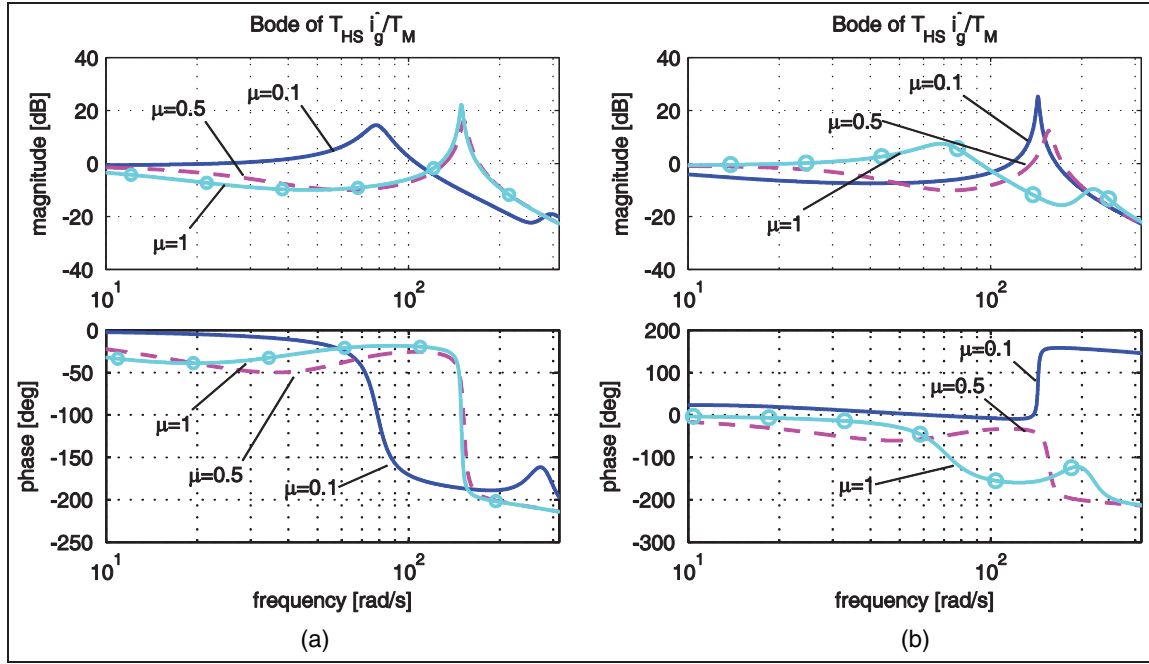


Figure 3. Bode plot (magnitude and phase) of the adimensionalized half-shaft torque for an input torque request with longitudinal slip ratios σ of (a) 0.05 and (b) $\sigma_{PEAK} = 0.01$, at different values of the tyre–road friction coefficient μ ($l_{HS} = 506$ mm; vehicle speed, 50 km/h).

response of the system in the case of a low-friction surface with snow, which gives rise to a monotonically increasing force characteristic of the tyres as a function of the slip ratio.

In conclusion, the case-study individual wheel powertrain is characterised by a higher value of the first natural frequency than the powertrain of an internal-combustion-engine-driven vehicle, which results in an improvement in the vehicle driveability. However, the open-loop dynamic response of the powertrain, due to the torsional dynamics of the half-shafts, is not satisfactory for an ABS and TC based on direct slip control. For example, Johansen et al.¹² mentioned a bandwidth of 72 rad/s (about 11.5 Hz) for the electromechanical friction brake callipers adopted in that specific vehicle application.

Control system design in the frequency domain

Figure 4 is the linearized block diagram of TC based on slip ratio feedback control without any compensation of the half-shaft torque dynamics, similar to the system presented by Hori et al.²⁷ The estimation of the slip ratio implies the availability of precise information on the longitudinal speed of a vehicle, which can be easily derived, for the case of a two-wheel-drive vehicle, from the wheel speed sensors of the undriven wheels, which would be adopted in any case for the purpose of anti-lock braking control. This method for estimation of the vehicle speed is reasonably reliable also in the case of cornering conditions. The main non-linearities

of the control system are represented by the switch for the activation and deactivation conditions based on two different slip ratio thresholds, and by the fact that during TC activation the algorithm selects the lower value between the driver torque demand and the output torque from the TC system.

In the frequency domain, for a vehicle with a TC system based on direct slip control implemented through a proportional–integral–derivative (PID) controller, the slip ratio $\bar{\sigma}(s)$ of the tyres is given by

$$\bar{\sigma}(s) = \frac{[\bar{\sigma}_{ref}(s)K_{PID,TC}(s) + \bar{T}_{M,ref}(s)]G_{PLANT}(s)}{1 + K_{PID,TC}(s)G_{PLANT}(s)} \quad (10)$$

The slip ratio is due to the combination of the reference torque contribution $\bar{T}_{M,ref}(s)$ imposed by the driver and the torque correction calculated by the traction controller. For a first approximation design of the

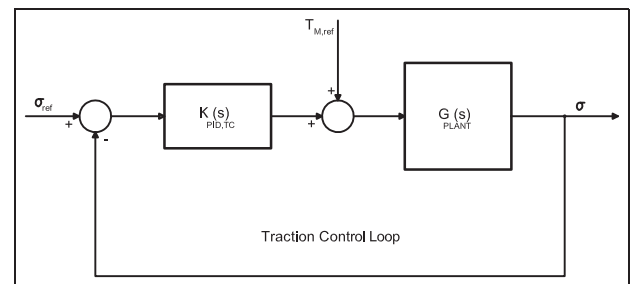


Figure 4. Block diagram of conventional TC based on direct slip control.

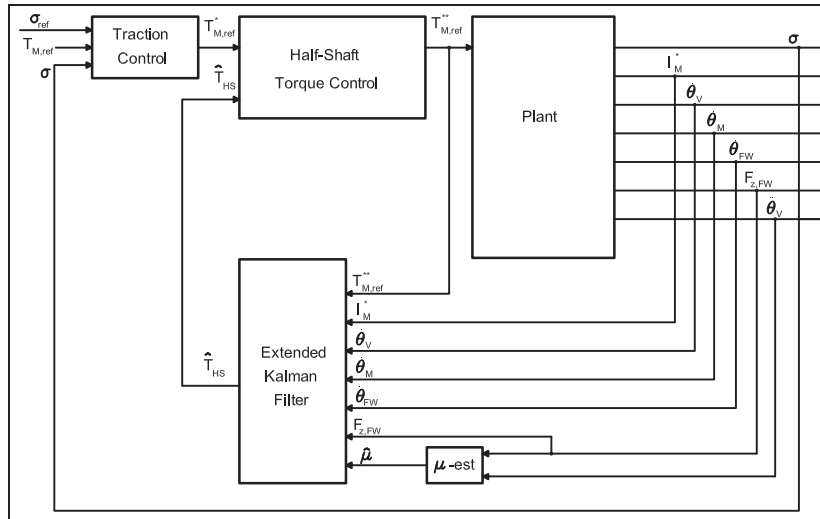


Figure 5. Block diagram of TC based on direct slip control as the outer control loop and the feedback controller for the compensation of the half-shaft torque dynamics as the inner control loop, including the schematic diagram detailing the input and output signals of the extended Kalman filter for estimation of the half-shaft torque.

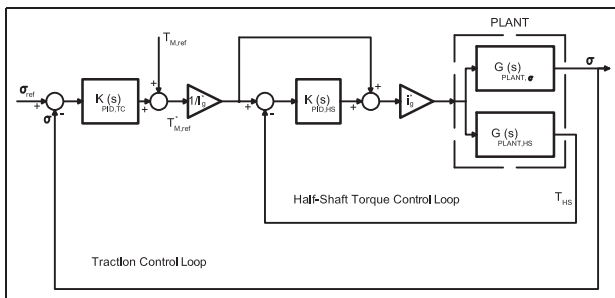


Figure 6. Block diagram of the linearized TC (used for control system design) based on direct slip control as the outer loop and the feedback controller for compensation of the half-shaft torque dynamics as the inner loop.

tracking performance of the traction controller, it is possible to consider the closed-loop transfer function of the control system, defined by the loop gain transfer function $K_{PID,TC}(s)G_{PLANT}(s)$, according to

$$G_{CL,TC,conventional}(s) = \frac{\bar{\sigma}}{\bar{\sigma}_{ref}}(s) = \frac{K_{PID,TC}(s)G_{PLANT}(s)}{1 + K_{PID,TC}(s)G_{PLANT}(s)} \quad (11)$$

In equation (10) the term $\bar{T}_{M,ref}(s)$ can be ignored, as it can be considered a disturbance during a manoeuvre with TC intervention. For a second approximation design of the controller, the analysis of the robustness of the control system against the variation in the driver torque demand during TC interventions should be carried out (but is not attempted in this paper).

An ABS and TC based on direct slip control both need the precise actuation of the wheel braking and traction torques. These are characterised by significant dynamics (already outlined in the second section) if the reference motor torque for achieving a target wheel torque is purely calculated by the ABS or TC as the desired wheel torque multiplied by the transmission gear ratio (less than 1 according to the conventions adopted in the article).

The idea of this research is the implementation of an inner control loop for the feedback compensation of the torque oscillations at the wheels (Figure 5). The feedforward component of the reference torque at the half-shaft is computed as the reference electric motor torque $\bar{T}_{M,ref}^*$, depending on the driver input and the TC, divided by i_g^* .

A PID controller was implemented for feedback control of the half-shaft torque as the inner control loop,

Table 1. Gain margin values for different tunings of the proportional gain of the TC system.

		Gain margin value (dB) for the following tunings of the proportional gain				
		10 N m	50 N m	100 N m	300 N m	700 N m
Without half-shaft torque control	20	-6.34	-20.3	-26.3	-35.9	-43.2
	50	7.82	-2.16	-12.2	-21.7	-29.1
	90	15.2	1.26	-4.76	-13.3	-21.6
With half-shaft torque control	20	36.6	22.7	16.6	7.1	-0.27
	50	44.6	30.6	24.6	15.1	7.7
	90	49.8	35.8	29.8	2.2	12.9

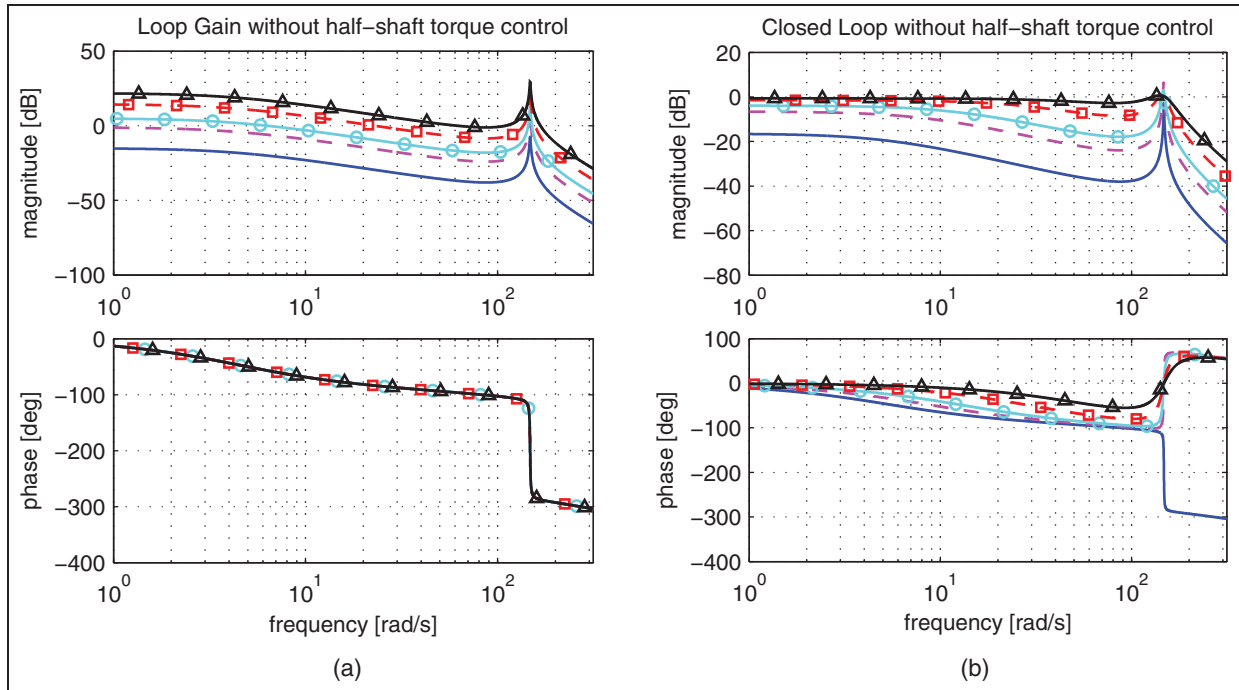


Figure 7. Bode plots of (a) the loop gain transfer function and (b) the closed-loop transfer function of the vehicle including TC and excluding half-shaft torque control. The frequency response plots are shown for different values of the proportional gain of the traction controller: —, 10 N m; ---, 50 N m; ○, 100 N m; □, 300 N m; △, 700 N m.

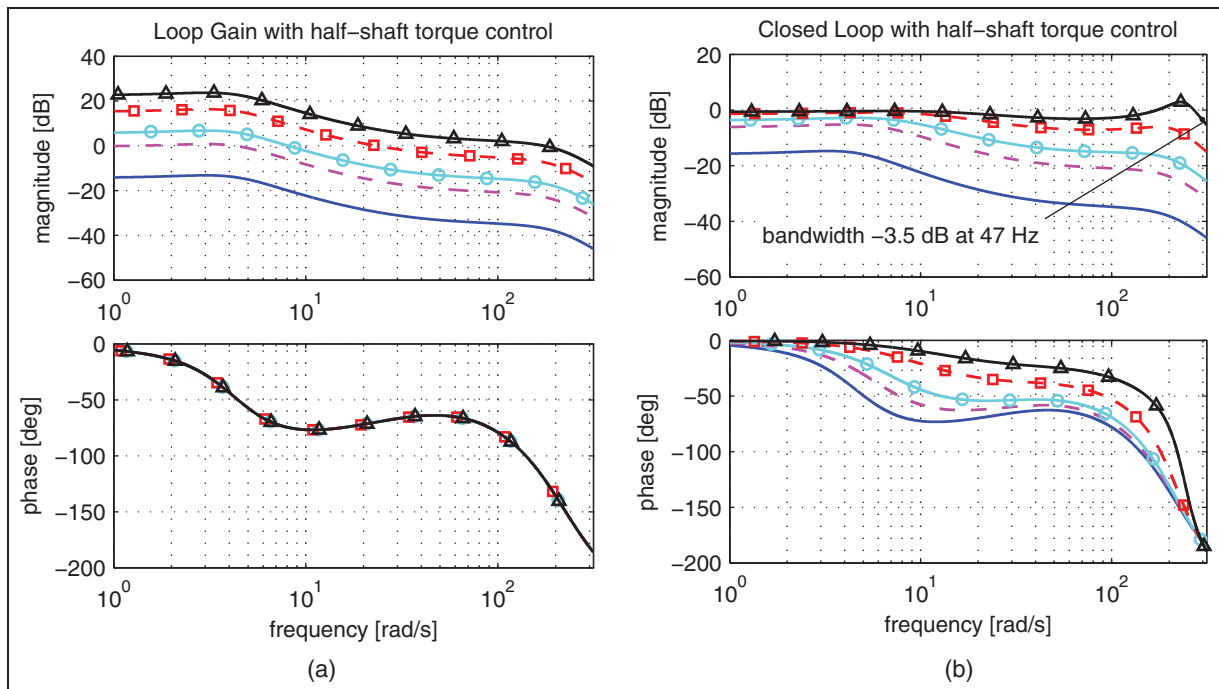


Figure 8. Bode plots of (a) the loop gain transfer function and (b) the closed-loop transfer function of the vehicle including TC and half-shaft torque control. The frequency response plots are shown for different values of the proportional gain of the traction controller: —, 10 N m; ---, 50 N m; ○, 100 N m; □, 300 N m; △, 700 N m.

according to the linearized block diagram of the system in Figure 6. The gains were designed for the achievement of a gain margin of about 7 dB and a phase margin of 60°, at a vehicle speed of 50 km/h.

As the half-shaft torque cannot be directly measured on the plant, the system requires the implementation of an extended Kalman filter (Figure 5), for estimation of the half-shaft torque level. The extended Kalman filter

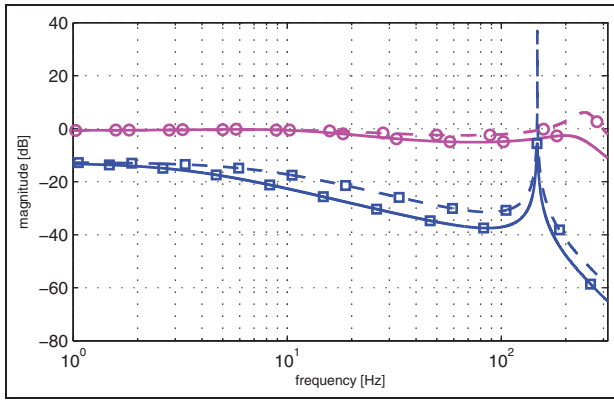


Figure 9. Bode magnitude plot of the closed-loop transfer function of the system with TC control, excluding the inner loop ($\square$) and including the inner loop ($\circ$), at two different values of the vehicle speed: ---, 40 km/h; —, 80 km/h. The proportional gain of the TC is equal to 17 N m in the system without the half-shaft torque control loop and it is 700 N m in the system with the inner loop.

is based on the non-linear electric powertrain model presented in the second section, with the exclusion of the relaxation length of the tyres and the equations of the longitudinal dynamics of the vehicle. Appendix 4 presents the structure and the details of the formulation of the extended Kalman filter adopted in this contribution. The measured values of the angular speeds of the motor and the wheels and the measurement of the motor torque were provided through sensor simulation and were pre-conditioned with properly tuned low-pass filters. Selection of the noise level for the measurements was carried out consistently with the experimental values reported by Johansen et al.¹² This Kalman filter can also provide a reliable estimation of the longitudinal force of the tyres, being characterised by a non-linear tyre model. An alternative implementation of a similar Kalman filter was presented by Amann et al.,²³ where the filter is simplified from the viewpoint of the estimation of the force of the tyres but includes, within its non-linear model, the simulation of transmission plays, which can provoke significant driveline oscillations. The best option for the detailed implementation of the Kalman filter will be experimentally verified by the present authors during the vehicle-testing activity which is going to be implemented in the future.

A comprehensive set of simulations was carried out in order to check the robustness of the extended Kalman filter estimation against the variation in the tyre-road friction coefficient, and against the variation in the tyre parameters. In particular, for the specific case study, the filter estimation is not significantly affected even by an incorrect estimation of the friction coefficient of the tyres, when the slip ratio is not close to the value corresponding to the peak longitudinal force. Therefore, within the implementation, the extended Kalman filter utilises a fairly crudely

estimated value of the actual friction of the tyres, based on the methodology presented by Hori et al.²⁷

The new feedback controller is useful during normal driving conditions for improving vehicle driveability, and during TC or ABS actuation for improving wheel dynamics. If we ignore the dynamics induced by the presence of the Kalman filter on the feedback loop, the loop gain transfer function for the system including the new control loop is

$$G_{LG}(s) = G_{PLANT,\sigma} K_{HS,CL}(s) K_{PID,TC}(s) \\ = G_{PLANT,\sigma} \left[\frac{1 + K_{PID,HS}(s)}{1 + i_g^* K_{PID,HS}(s) G_{PLANT,HS}(s)} \right] K_{PID,TC}(s) \quad (12)$$

At the moment, within this research focused on the analysis of the in-board powertrain dynamics, the TC system does not include a robust adjustment of σ_{ref} as a function of the estimated characteristics of the tyres. As the focus of this article is on the effect of the half-shaft on the dynamics of the individual wheel powertrain, the traction controllers in Figures 4 and 6 were implemented as basic proportional controllers. The design of the gains for the TC systems were based on the analysis of the gain margin and system stability in the Bode diagrams of the loop gain transfer functions. Typical desired values of the gain margin are between 3 dB and 10 dB.

Figures 7 and 8 compare the performance of the TC systems of Figures 4 and 6, from the viewpoint of the Bode plots of their loop gain (Figures 7(a) and 8(a)) and closed-loop (Figures 7(b) and 8(b)) transfer functions. The results were obtained through linearization of the model in an operating condition close to the peak point of the longitudinal force characteristic of the tyres (precisely at a slip ratio with an offset of 0.01 to the left of the peak slip ratio), at a vehicle speed of 50 km/h. The adoption of the half-shaft torque control inner loop provides a significant enhancement of system stability. This allows selection of higher values of the proportional gain of the TC, resulting in a quicker dynamic response of the controller, for the same values of gain margin, which are compared in Table 1 for different tunings of the proportional gain of the TC. Table 1 also shows a significant increase in the gain margin as a function of the vehicle speed, which implies the requirement for a gain scheduling of the controller. Owing to the high non-linearity of the system, the fine tuning of the controller requires simulations in the time domain, which are presented in the fourth section. Figures 7(b), 8(b) and 9 compare the closed-loop performance of the control systems of Figures 4 and 6 in the frequency domain. In particular, with the system including the feedback controller on the half-shaft torque, it is possible to achieve an acceptable tracking capability (−3.5 dB of closed-loop response) up to a frequency of about 47 Hz, well beyond the expected frequency range for TC and ABS. Additional benefits should be achievable by adopting more advanced controllers for the TC function, such as compensators and

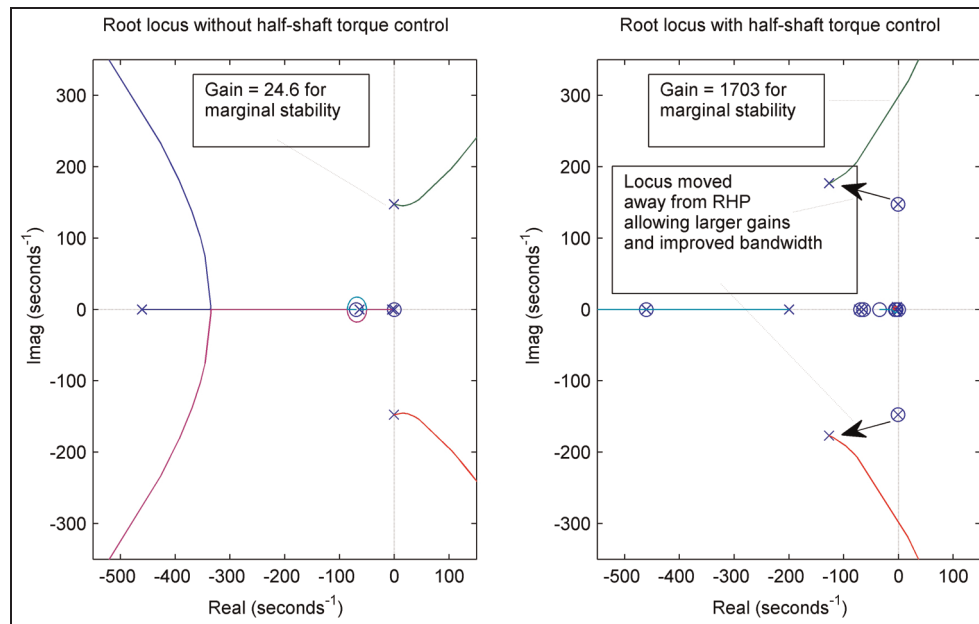


Figure 10. Root locus analysis excluding and including half-shaft torque control at 50 km/h. RHP: right-hand plane.

linear quadratic regulators, which will be the object of future research.

Figure 10 is the root locus plot for the overall closed-loop system, excluding (Figure 10(a)) and including (Figure 10(b)) the half-shaft torque control system. The adoption of the novel half-shaft torque control system cancels the poles near the imaginary axis and moves the part of the root locus which progresses towards the right-hand plane zeros (not pictured) further to the left. This significantly contributes to system stability, allowing the adoption of higher values of the proportional gain of the TC system.

Results in the time domain

The half-shaft torque inner control loop presented in the previous section is a useful method for modifying the loop gain frequency response such that TC or ABS slip (or anti-jerk) control design is much simpler and provides a higher performance.

Figure 11 demonstrates the functionality of the novel controller through simulation in the time domain. The electric motor torque (and hence the reference wheel torque) is ramped up to a desired steady state level, and then a sinusoidal oscillation at a frequency of 9 Hz is superimposed. In Figure 11(a), the vehicle equipped with the control system can follow the reference, while the vehicle without the feedback control system on the half-shaft torque suffers from a mean level offset, a phase lag and a significant amplitude error in the oscillations.

Simulations of tip-in tests (Figure 12) with the nonlinear model show typical oscillations of the acceleration profile of the vehicle induced by the torsional

dynamics of the electric powertrains. The vehicle equipped with the feedback half-shaft torque controller is characterised by an appreciably smoother acceleration profile, with attenuated oscillations in comparison with the vehicle without the novel control system, and so the controller can be considered as an indirect anti-jerk controller.

Finally, Figure 13 shows the comparison of the performance of the passive vehicle, the vehicle equipped with the TC in Figure 4 (according to two different tunings) and the vehicle equipped with the TC and the half-shaft torque control system in Figure 6, during a tip-in test from an initial speed of 50 km/h. The wheel torque level during the first part of the tip-in manoeuvre is beyond the friction limit of the tyres, which implies significant wheel spinning for both the passive vehicle and the vehicle with TC only (proportional control). The TC without half-shaft torque control requires a significantly low value of the proportional gain of the controller to maintain system stability, as discussed in the third section. The stand-alone TC tuned according to the system frequency response characteristics was unable to achieve the desired performance level with a proportional gain only. Therefore, a PID control system was also implemented to replace the TC gain controller (still without the inner half-shaft torque control loop) in order to reduce the slip ratio especially during the first part of the wheel-spinning phase.

In frequency domain terms, the PID controller can be utilised as a band-stop compensator to reduce the loop gain around the first resonant peak in order to improve the gain margin. In any case, even this TC configuration achieves more irregular slip control dynamics than the system including the half-shaft torque inner control loop. The vehicle equipped with the novel

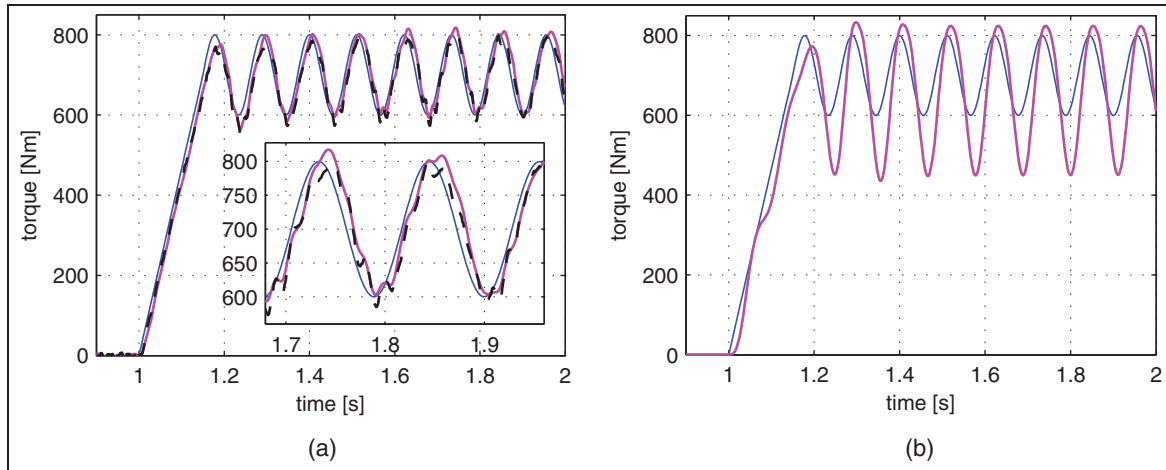


Figure 11. The half-shaft reference torque (—), the actual half-shaft torque (—) and Kalman filter estimation of the half-shaft torque (---) (a) with half-shaft torque control and (b) with no control. The electric motor drive reference torque (in the figure, the half-shaft torques are plotted) is a fast ramp up to 70 N m plus a sinusoidal oscillation with an amplitude of 10 N m and a frequency of 9 Hz. The initial speed of the vehicle is 50 km/h.

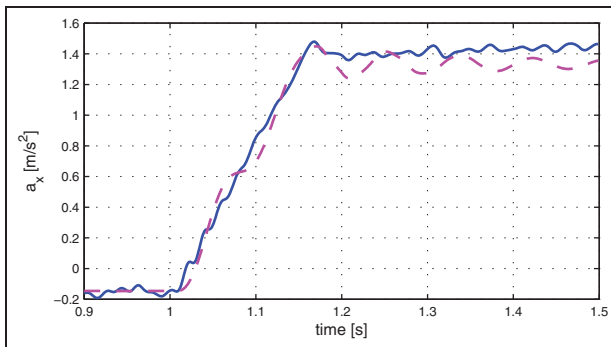


Figure 12. Longitudinal acceleration profile of the vehicle during a tip-in manoeuvre with half-shaft torque control (—) and without half-shaft torque control (---). The final torque demand is 75 N m per electric motor starting from zero. The initial speed of the vehicle is 20 km/h.

controller and a basic proportional TC is characterised by the absence of any significant wheel spinning (a maximum slip ratio of about 0.16) and a higher speed profile than those of the other three vehicles.

The main performance metrics relating to the manoeuvre of Figure 13 are reported in Table 2. In particular, the novel controller proposed in this contribution allows the achievement of an average value of slip ratio

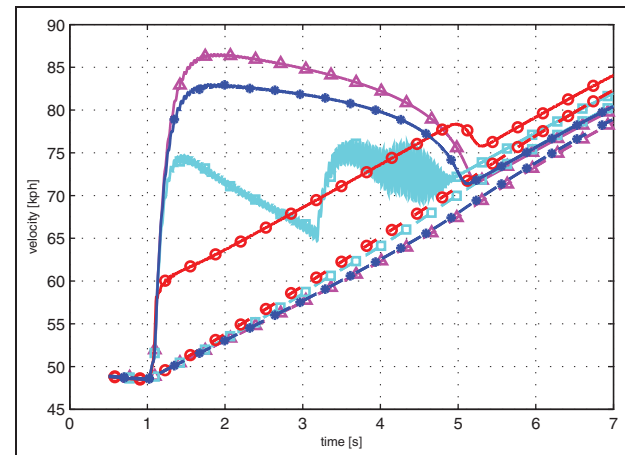


Figure 13. The speeds of the wheels (—) and the vehicle (---) during a tip-in manoeuvre: Δ , vehicle without the control system; *, vehicle with TC based on proportional gain; $\square$, vehicle with TC based on a PID controller; $\circ$, vehicle with TC based on proportional gain and the novel half-shaft torque control.

very close to the reference value, with a reduction in the peak values of the longitudinal slip ratio by a percentage between 85% and 116% in comparison with the vehicles equipped with TC systems without the half-shaft torque feedback loop.

Table 2. Performance metrics of the different TC solutions presented in the paper, during the same manoeuvre as in Figure 13.

	No control	Proportional gain TC	PID TC	Proportional gain TC with half-shaft torque control
Average acceleration	1.36 m/s ²	1.39 m/s ²	1.51 m/s ²	1.58 m/s ²
Average slip ratio ^a	0.262 (+ 113%)	0.239 (+ 94%)	0.152 (+ 24%)	0.123
Peak slip ratio ^a	0.403 (+ 129%)	0.38 (+ 116%)	0.325 (+ 85%)	0.176

PID: proportional–integral–derivative; TC: traction control.

^aThe percentage variations are referred to the proportional TC system including half-shaft torque control (configuration in the last column).

Conclusion

The article has demonstrated the potential effects induced by the torsional dynamics of the half-shafts on the response of individual wheel in-board powertrains. In particular, the first natural frequency of the system, which is due to the torsional stiffness of the half-shaft, the mass moments of inertia of the powertrain components and the tyre characteristics at the operating point, is responsible for a significant deterioration in the system response in terms of the wheel torque. The in-board powertrain dynamics were compensated through a feedback controller, based on the half-shaft torque estimation using an extended Kalman filter. The performance of a case-study TC system is significantly enhanced by the adoption of the designed control system, notably achieving a tracking bandwidth of up to 47 Hz at -3.5 dB.

Future work will consider incorporating this methodology into a general four-wheel-drive electric vehicle dynamics controller as part of the Electric Vehicle Control of Individual Wheel Torque for On- and Off-Road Conditions (E-VECTOORC) research activity of the Seventh Framework Programme of the European Commission.

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Appendix I

Notation

a_{ij}	element on the i th row and j th column of the state matrix	f_2	term of the rolling resistance coefficient of the tyre which is to be multiplied by the tangential speed of the wheel squared
a_{ij}^{KF}	element on the i th row and j th column of the matrix A^{KF}	$F_{z,W}$	vertical force between the wheel and the road
A	state matrix (for the state-space model formulation)	g	acceleration due to gravity
A^{KF}	Jacobian matrix of the partial derivatives of f with respect to X	$G_{CL,TC,conventional}$	closed-loop transfer function $(\bar{\sigma}/\bar{\sigma}_{ref})(s)$ of the vehicle with the traction control system based on direct slip control, without feedback control of the half-shaft torque
b	rear semi-wheelbase	G_{LG}	loop gain transfer function of the system
B	input matrix (for the state-space model formulation)	G_{PLANT}	transfer function of the plant (the vehicle)
C	output matrix (for the state-space model formulation)	$G_{PLANT,HS}$	transfer function of the plant (the vehicle), providing the half-shaft torque as the output
C_d	aerodynamic drag coefficient	$G_{PLANT,\sigma}$	transfer function of the plant (the vehicle), providing the slip ratio as the output
C_l	aerodynamic lift coefficient	h	Kalman filter non-linear measurement equation
C_m	aerodynamic pitch moment coefficient	h_{CG}	height of the centre of gravity
d_{HS}	half-shaft diameter	H	Jacobian matrix of the function h with respect to X
D	feedthrough matrix (for the state-space model formulation)	i_g^*	overall gear ratio of the single-speed transmission
f	Kalman filter non-linear difference equation of the system state	i_{g1}	gear ratio of the first stage of the single-speed transmission
f_r	rolling resistance coefficient of the tyre	i_{g2}	gear ratio of the second stage of the single-speed transmission
f_0	part of the rolling resistance coefficient of the tyre which is independent of the tangential speed of the wheel	i_{g3}	gear ratio of the third stage of the single-speed transmission
f_1	term of the rolling resistance coefficient of the tyre which is to be multiplied by the tangential speed of the wheel	J_{HS}	mass moment of inertia of the half-shaft
		J_M	mass moment of inertia of the electric motor
		J_W	mass moment of inertia of a wheel
		J_M^*	mass moment of inertia of the electric powertrain at the motor
		J_V^*	equivalent mass moment of inertia of the vehicle
		J_W^*	equivalent mass moment of inertia of a wheel
		J_1	mass moment of inertia of the transmission input shaft (Figure 1)
		J_2	mass moment of inertia of transmission shaft 2 (Figure 1)
		J_3	mass moment of inertia of transmission shaft 3 (Figure 1)
		J_4	mass moment of inertia of the transmission output shaft (Figure 1)
		k	discretisation step parameter
		k_{HS}	torsional stiffness of the half-shaft
		K	Kalman gain

$K_{HS,CL}$	closed-loop transfer function of the half-shaft torque control system	Δt_{KF}	Kalman filter time step
$K_{PID,TC}$	transfer function of the traction control system (proportional–integral–derivative controller)	η^*	overall efficiency of the single-speed transmission including the constant-speed joint
l_{HS}	half-shaft length	η_{j1}	efficiency of the inner constant-speed joint
L	wheelbase of the vehicle	η_{j2}	efficiency of the outer constant-speed joint
L_{tyre}	relaxation length of the tyre	η_1	efficiency of the first stage of the single-speed transmission
m	mass of the vehicle	η_2	efficiency of the second stage of the single-speed transmission
P	error covariance matrix	η_3	efficiency of the third stage of the single-speed transmission
$P_{M,max}$	maximum motor power	$\theta_M, \dot{\theta}_M, \ddot{\theta}_M$	angular displacement, speed and acceleration respectively of the electric motor shaft
Q	process noise covariance matrix	$\dot{\theta}_V, \ddot{\theta}_V$	angular speed and acceleration respectively of the equivalent inertia of the vehicle
R	measurement noise covariance	$\theta_W, \dot{\theta}_W, \ddot{\theta}_W$	angular displacement, speed and acceleration respectively of a driven wheel
R_W	radius of the tyre	ρ	density of air
s	Laplace variable	σ	slip ratio of the tyre
S	frontal area of the vehicle (for computation of the aerodynamic force and the moment)	σ_{PEAK}	slip ratio of the tyre at which the peak longitudinal force is achieved
$T_{d,W}$	delayed (because of the effect of the relaxation length of the tyre) torque between tyre and road (with the time derivative $\dot{T}_{d,FW}$)	σ_{ref}	reference slip ratio
T_{HS}	half-shaft torque	τ_M	electrical time constant of the electric motor drive
T_M	electric motor torque (with the time derivative $\dot{T}_M$)	μ	tyre–road friction coefficient
$T_{M,max}$	maximum electric motor torque	Superscripts $\hat{}$ estimated variable $-$ Laplace transform of a variable in the time domain (overbar) $\sim$ measured state	
$T_{M,ref}$	reference electric motor torque, according to the driver input		
$T_{windage}$	electric motor torque windage losses		
T_W	net torque transmitted by the half-shaft to the driven wheel		
$T_{W,Pacejka}$	tyre torque according to the Pacejka magic formula model	Subscripts F front component of the vehicle R rear component of the vehicle 0 initial condition of a state variable	
$T_{M,ref}^*$	reference electric motor torque from traction control (Figure 5)		
$T_{M,ref}^{**}$	reference electric motor torque from half-shaft torque control (Figure 5)		
u_W	longitudinal offset (due to the rolling resistance coefficient) between the normal force of the tyre and the centre of the contact patch		
U	input vector (for the state-space model formulation)	Appendix 2 Matrices of the state-space formulation of the linearised vehicle model Each element of the matrices of the first equation of equation (9) is defined here. The <i>state vector</i> is	
V	measurement noise vector		
V^{KF}	Jacobian matrix of the function h with respect to V		
W	process noise vector		
W^{KF}	Jacobian matrix of f with respect to W	$X^T = [\dot{\theta}_M, \dot{\theta}_{FW}, \dot{\theta}_V, \theta_M, \theta_{FW}, T_M, T_d]$ (13)	
X	state vector (for the state-space model formulation)		
Y	output vector (for the state-space model formulation)		
Z	measurement vector		
β_{HS}	torsional damping coefficient of the half-shaft	The <i>state matrix</i> is	

$$A = \begin{bmatrix} a_{1,1} & a_{1,2} & 0 & a_{1,4} & a_{1,5} & a_{1,6} & 0 \\ a_{2,1} & a_{2,2} & 0 & a_{2,4} & a_{2,5} & 0 & a_{2,7} \\ 0 & 0 & a_{3,3} & 0 & 0 & 0 & a_{3,7} \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & a_{6,6} & 0 \\ 0 & a_{7,2} & a_{7,3} & 0 & 0 & 0 & a_{7,7} \end{bmatrix} \quad (14)$$

where the non-null terms are

$$\begin{aligned} a_{1,1} &= -\frac{i_g^{*2} \beta_{HS}}{\eta^* J_M^*}, a_{1,2} = \frac{i_g^* \beta_{HS}}{\eta^* J_M^*}, a_{1,4} = -\frac{i_g^{*2} k_{HS}}{\eta^* J_M^*}, \\ a_{1,5} &= \frac{i_g^* k_{HS}}{\eta^* J_M^*}, a_{1,6} = \frac{1}{J_M^*} \\ a_{2,1} &= -\frac{\eta_{j2} \beta_{HS} i_g^*}{J_{FW}^*}, \\ a_{2,2} &= -\frac{\eta_{j2} \beta_{HS} + (f_1 + 2f_2 R_W \dot{\theta}_{FW,0}) R_W^2 F_{z,FW,0}}{J_{FW}^*} \\ a_{2,4} &= \frac{\eta_{j2} k_{HS} i_g^*}{J_{FW}^*}, a_{2,5} = -\frac{\eta_{j2} k_{HS}}{J_{FW}^*}, a_{2,7} = -\frac{1}{J_{FW}^*} \\ a_{3,3} &= -\frac{2(f_1 + 2f_2 R_W \dot{\theta}_{V,0}) R_W^2 F_{z,RW} + \rho C_d S R_W^3 \dot{\theta}_{V,0}}{J_V^*}, \\ a_{3,7} &= \frac{2}{J_V^*} \\ a_{6,6} &= -\frac{1}{\tau_M} \\ a_{7,2} &= \frac{\dot{\theta}_{V,0} \beta_{FW}}{\dot{\theta}_{FW,0} \tau_W}, a_{7,3} = -\frac{\beta_{FW}}{\tau_W}, a_{7,7} = -\frac{1}{\tau_W} \\ J_M^* &= (J_M + J_1) + \frac{i_{g1}^2}{\eta_1} J_2 + \frac{(i_{g1} i_{g2})^2}{\eta_1 \eta_2} J_3 + \frac{(i_{g1} i_{g2} i_{g3})^2}{\eta_1 \eta_2 \eta_3} \\ &\quad \left(J_4 + \frac{1}{2} \frac{J_{HS}}{\eta_{j1}} \right) \\ J_{FW}^* &= J_{FW} + \eta_{j2} \frac{J_{HS}}{2}, J_V^* = 2J_{RW} + mR_W^2 \\ i_g^* &= i_{g1} i_{g2} i_{g3}, \eta^* = \eta_1 \eta_2 \eta_3 \eta_{j1} \end{aligned}$$

The input vector is

$$U^T = \left[T_{ref}, -f_0 R_W F_{z,FW,0}, f_2 R_W^3 \dot{\theta}_{FW,0}^2 F_{z,FW,0}, -f_0 R_W F_{z,RW,0}, \right. \\ \left. f_2 R_W^3 \dot{\theta}_{V,0}^2 F_{z,RW,0}, \frac{1}{2} \rho C_d S R_W^3 \dot{\theta}_{V,0}^2 \beta_{FW} (\dot{\theta}_{FW,0} - \dot{\theta}_{V,0}) \right] \quad (15)$$

The input matrix is

$$B = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{J_{FW}^*} & \frac{1}{J_{FW}^*} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{2}{J_V^*} & \frac{2}{J_V^*} & \frac{1}{J_V^*} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{1}{\tau_M} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\dot{\theta}_{W,0} R_W}{L_{lyre}} \end{bmatrix} \quad (16)$$

Appendix 3

Main parameters of the vehicle

A list of the main parameters of the vehicle is given in Table 3.

Table 3. Main parameters of the vehicle.

Symbol	Value
<i>Vehicle parameters</i>	
M	2500 kg
L	2.66 m
b	1.35 m
h_{CG}	0.65 m
C_d	0.39
S	2.76 m ²
<i>Wheel parameters</i>	
R_W	370 mm
$J_{FW} = J_{RW}$	0.9 kg m ²
<i>Half-shaft parameters</i>	
Material	Steel
d_{HS}	30 mm
l_{HS}	506 mm
<i>Gearbox parameters</i>	
i_g^*	1/10
<i>Motor parameters</i>	
$T_{M,max}$	200 N m
$P_{M,max}$	80 kW
J_M	0.016 kg m ²

Appendix 4

Formulation of the extended Kalman filter for half-shaft torque estimation

The process is governed by the non-linear stochastic difference equation

$$X_k = f(X_{k-1}, U_{k-1}, W_{k-1}) \quad (17)$$

where W is the process noise vector. The vector of the measured variables is

$$Z_k = h(X_k, V_k) \quad (18)$$

where h is the non-linear measurement equation and V is the measurement noise vector.

The extended Kalman filter has been implemented according to the general algorithm detailed in the following steps.

Step 1: Provide an initial estimate for X_{k-1} and P_{k-1} , where P is the error covariance matrix.

Step 2: Project the state ahead, which is given by

$$\hat{X}_k^{(-)} = f(\hat{X}_{k-1}, U_{k-1}, 0) \quad (19)$$

where $\hat{X}_k^{(-)}$ is the a-priori estimate of the state at step k , $\hat{X}_{k-1}$ is the a-posteriori estimate of the state at step $k-1$, U_{k-1} is the input vector at step $k-1$ and 0 is the process error estimate at step $k-1$.

Step 3: Project the error covariance ahead, which is given by

$$P_k^{(-)} = A_k^{KF} P_{k-1} A_k^{KFT} + W^{KF} Q_{k-1} W_k^{KFT} \quad (20)$$

where A^{KF} is the Jacobian matrix of the partial derivatives of f with respect to X , W^{KF} is the Jacobian matrix of f with respect to W and Q is the process noise covariance.

Step 4: Compute the Kalman gain

$$K_k = P_k^- H_k^T (H_k P_k^- H_k^T + V_k^{KF} R_k V_k^{KFT})^{-1} \quad (21)$$

where H is the Jacobian matrix of h with respect to X , V^{KF} is the Jacobian matrix of h with respect to V and R is the measurement noise covariance.

Step 5: Update the estimate with measurement Z_k to give

$$\hat{X}_k = \hat{X}_k^- + K_k [Z_k - h(\hat{X}_k^-, 0)] \quad (22)$$

Step 6: Update the error covariance to give

$$P_k = (I - K_k H_k) P_k^- \quad (23)$$

For the particular system of this contribution, the state vector at step k is

$$X_k = \{\dot{\theta}_{Mk}, \dot{\theta}_{Wk}, \theta_{Mk}, \theta_{Wk}, T_{Mk}\}^T \quad (24)$$

The non-linear system of difference equations $X_k = f(X_{k-1}, U_{k-1}, W_{k-1})$ is

$$\begin{aligned} \dot{\theta}_{Mk} = & \left(1 - \frac{i_g^{*2} \beta_{HS} \Delta t_{KF}}{\eta^* J_M^*}\right) \dot{\theta}_{Mk-1} \\ & + \frac{i_g^* \beta_{HS} \Delta t_{KF}}{\eta^* J_M^*} \dot{\theta}_{Wk-1} - \frac{i_g^{*2} k_{HS} \Delta t_{KF}}{\eta^* J_M^*} \theta_{Mk-1} \\ & + \frac{i_g^* k_{HS} \Delta t_{KF}}{\eta^* J_M^*} \theta_{Wk-1} + \frac{\Delta t_{HS}}{J_M^*} T_M + w_{1k-1} \end{aligned} \quad (25)$$

$$\begin{aligned} \dot{\theta}_{Wk} = & \frac{\eta_{j2} i_g^* \beta_{HS} \Delta t_{KF}}{J_W^*} \dot{\theta}_{Mk-1} \\ & + \left(1 - \frac{\eta_{j2} \beta_{HS} \Delta t_{KF}}{J_W^*} - \frac{f_1 R_W^2 F_{z,W} \Delta t_{KF}}{J_W^*} \right. \\ & \quad \left. - \frac{f_2 R_W^3 F_{z,W} \Delta t_{KF}}{J_W^*} \right) \dot{\theta}_{Wk-1} \\ & + \frac{\eta_{j2} i_g^* k_{HS} \Delta t_{KF}}{J_W^*} \theta_{Mk-1} - \frac{\eta_{j2} k_{HS} \Delta t_{KF}}{J_W^*} \theta_{Wk-1} \\ & - \frac{T_{W, Pacejka_{k-1}} \Delta t_{KF}}{J_W^*} - \frac{f_0 F_{z,W} R_W \Delta t_{KF}}{J_W^*} + w_{2k-1} \end{aligned} \quad (26)$$

$$\theta_{Mk} = \dot{\theta}_{Mk-1} \Delta t_{KF} + \theta_{Mk-1} + w_{3k-1} \quad (27)$$

$$\theta_{Wk} = \dot{\theta}_{Wk-1} \Delta t_{KF} + \theta_{Wk-1} + w_{4k-1} \quad (28)$$

$$T_{Mk} = \left(1 - \frac{\Delta t_{KF}}{\tau_M}\right) T_{Mk-1} + T_{M, refk-1}^* \frac{\Delta t_{KF}}{\tau_M} + w_{5k-1} \quad (29)$$

The measurement vector is $\tilde{Z}_k = \{\tilde{\theta}_{Mk}, \tilde{\theta}_{Wk}, \tilde{T}_{Mk}\}^T$ and its elements at step k are

$$\tilde{\theta}_{Mk} = \dot{\theta}_{Mk} + v_{1k} \quad (30)$$

$$\tilde{\theta}_{Wk} = \dot{\theta}_{Wk} + v_{2k} \quad (31)$$

$$\tilde{T}_{Mk} = T_{Mk} + v_{3k} \quad (32)$$

A^{KF} is the Jacobian matrix of partial derivatives of f with respect to X and is given by

$$A^{KF} = \begin{bmatrix} a_{11}^{KF} & a_{12}^{KF} & a_{13}^{KF} & a_{14}^{KF} & a_{15}^{KF} \\ a_{21}^{KF} & a_{22}^{KF} & a_{23}^{KF} & a_{24}^{KF} & 0 \\ a_{31}^{KF} & 0 & 1 & 0 & 0 \\ 0 & a_{42}^{KF} & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & a_{55}^{KF} \end{bmatrix} \quad (33)$$

$$a_{11}^{KF} = 1 - \frac{i_g^{*2} \beta_{HS} \Delta t_{KF}}{\eta^* J_M^*}$$

$$a_{12}^{KF} = \frac{i_g^* \beta_{HS} \Delta t_{KF}}{\eta^* J_M^*}$$

$$a_{13}^{KF} = -\frac{i_g^{*2} k_{HS} \Delta t_{KF}}{\eta^* J_M^*}$$

$$a_{14}^{KF} = \frac{i_g^* k_{HS} \Delta t_{KF}}{\eta^* J_M^*}$$

$$a_{15}^{KF} = \frac{\Delta t_{HS}}{J_M^*}$$

$$a_{21}^{KF} = \frac{\eta_{j2} i_g^* \beta_{HS} \Delta t_{KF}}{J_W^*}$$

$$a_{22}^{KF} = 1 - \frac{\eta_{j2} \beta_{HS} \Delta t_{KF}}{J_W^*} - \frac{f_1 R_W^2 F_{z,W} \Delta t_{KF}}{J_W^*}$$

$$- \frac{2f_2 R_W^3 F_{z,W} \Delta t_{KF}}{J_W^*} \dot{\theta}_W - \frac{\partial T_{W, Pacejka}}{\partial \dot{\theta}_W} \frac{\Delta t_{KF}}{J_W^*}$$

$$a_{23}^{KF} = \frac{\eta_{j2} i_g^* k_{HS} \Delta t_{KF}}{J_W^*}$$

$$a_{24}^{KF} = -\frac{\eta_{j2} k_{HS} \Delta t_{KF}}{J_W^*}$$

$$a_{31}^{KF} = \Delta t_{KF}$$

$$a_{42}^{KF} = \Delta t_{KF}$$

$$a_{55}^{KF} = \left(1 - \frac{\Delta t_{KF}}{\tau_M}\right)$$

The Jacobian matrices H , W^{KF} and V^{KF} are identity matrices.